Waverly A. Chao

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MECHANICAL ENGINEERING
Cornell University · Class of 2029

EDUCATION

Cornell University | Ithaca, NY — Aug 2025 - Present
B.S. in Mechanical Engineering, expected May 2029 | Cornell Mars Rover Project Team

The Brearley School | New York, NY — Sep 2012 - Jun 2025
High School Diploma | Captain and Head of Build, FTC Robotics Team

ENGINEERING EXPERIENCE

Drives Team Member | Cornell Mars Rover — Nov 2025 - Present
- Contribute to the design and integration of the 2026-2027 rover suspension system as a member of the Drives subteam.
- Designed and implemented a panoramic camera mount and supported the drives electrical system during freshman year.

Machining Specialist | Cornell Space Systems Design Studio (SSDS) — Sep 2026 - Present
- Manual Machining of parts for the DeSCENT Flight Experiment, primarily lathe parts with some milling.
- DeSCENT builds a deployer for 100 small circuit boards that record atmospheric data as they fall back to the ground.

Team Captain and Head of Build | Brearley FTC Robotics Team — Sep 2021 - Jun 2025
- Led engineering design, prototyping, robot build, and match strategy through multiple winning seasons.
- Coordinated sponsorships, branding, technical documentation, and community outreach while teaching younger students.

Junior Researcher | NYU Tandon School of Engineering — Jun 2023 - Sep 2023
- Investigated optimization methods for lithium-ion half-cell batteries in the Transformative Materials and Devices Lab.
- Synthesized research findings and presented results at the conclusion of the program.

LEADERSHIP AND ADDITIONAL EXPERIENCE

Coach and Staff Member | Central Rock Gym Manhattan — May 2026 - Aug 2026
- Coached children ages 6-14 in climbing fundamentals, body and spatial awareness, and conditioning techniques.
- Supported front-desk, gym opening and closing procedures, specialized maintenance, cleaning, and facility operations.

Lead Camp Counselor | Brearley Summer Start — Jun 2025
- Designed curriculum and taught inaugural Brearley Robotics Summer Intensive for lower and middle school students.
- Led students through designing and building a competition-ready FTC robot in a single-week session.

Mechanical and Electrical Mentor | Youth Engineering and Entrepreneurship STEMinar — Sep 2023 - Jun 2025
- Mentored middle school students as they developed working prototypes addressing real-world problems.
- Concluded the year with Shark Tank-style presentations, emphasizing the physical and financial feasibility of their idea.

TECHNICAL SKILLS

Engineering | technical drawings, robotics, circuitry, electrical systems, PCB design, Microsoft Excel
Fabrication | Soldering, manual milling, lathe operation, 3D printing, printer repair, laser cutting, woodworking
CAD | Autodesk Fusion 360, Autodesk Inventor, Autodesk Vault, SolidWorks, Onshape, ANSYS (Finite Element Analysis)
Programming | Python, Java, Arduino, basic C++, Xcode, HTML, machine learning
Professional | Technical documentation, public speaking, mentoring, research, team leadership, conflict resolution